Deep Reinforcement Learning of UAV Tracking Control Under Wind Disturbances Environments

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***Abstract*— Aiming at the problems of strong nonlinearity, strong coupling, and unknown interference encountered in the flight control process of unmanned aerial vehicles (UAVs) in a complex dynamic environment and reinforcement-learning-based algorithm generalization, this study presents an innovative incremental reinforcement-learning-based algorithm for UAV tracking control in a dynamic environment. The main goal is to make a UAV able to adjust its control policy in a dynamic environment. The UAV tracking control task is transformed into a Markov decision process (MDP) and further investigated using an incremental reinforcement-learning-based method. First, a policy relief (PR) method is used to make UAVs capable of performing an appropriate exploration in a new environment. In this way, a UAV controller can mitigate the conflict between a new environment and the current knowledge to ensure better adaptability to a dynamic environment. In addition, a significance weighting (SW) method is developed to improve the utilization of episodes with higher importance and richer information. In the proposed method, learning episodes that include more useful information are assigned with higher importance weights. The numerical simulation, hardware-in-the-loop (HITL) experiments, and real-world flight experiments are conducted to evaluate the performance of the proposed method. The results demonstrate high accuracy and effectiveness and good robustness of the proposed control algorithm in a dynamic flight environment.**

***Index Terms*— Dynamic environment, reinforcement learning, tracking control, unmanned aerial vehicles (UAVs), wind disturbances.**

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I. Introduction

UNMANNED aerial vehicles (UAVs) have gained growing attention in the commercial and industrial fields owing to their characteristics of strong maneuverability, economical cost, and rapid development [1], [2], [3]. Also, due to these conveniences, UAVs have been widely applied in many applications, including traffic analysis, logistics and transportation, and resource exploration [4], [5], [6]. According to the differences in vehicle frame and flight mode, UAVs can be roughly divided into multirotor UAVs, fixed-wing UAVs, and flapping-wing aircraft [7], [8], [9]. Multirotor UAVs have the advantages of lightness, compactness, maneuverability, convenient take-off and landing, and an ability to realize aerial hovering and stable low-altitude flight, making them become one of the most widely applied aircraft.

To ensure UAV can achieve desired flight missions, a growing amounts of researches have been attracted to the implementation and application of robust UAV controller [4], [10], including proportional–integer–derivative (PID) control, the active disturbance rejection control (ADRC) control algorithm, the robust H_∞ control theory, and so on. In practical engineering, the traditional PID solution has been widely applied due to its robust and feasible characteristics. In [11], the gain scheduled PID algorithm for UAV attitude control was proposed; this algorithm can ensure a stable system state during the transition phase. In addition, Rosale et al. [12] designed a novel adaptive PID controller and applied an extended Kalman filter algorithm to filter the measurement errors, but its performance can degrade in a highly nonlinear control system. Certain improvements have been made to enhance the control accuracy. The controller proposed in [13] used a fuzzy mechanism to adjust controlled gains at various phases adaptively for speeding up the convergence. Yu et al. [14] proposed an adaptive sliding mode controller which has been designed to compensate for compound uncertainties from a nonlinear dynamic system. In [15], the Euler–Lagrange mechanical equation was used to derive and estimate the influences of UAV control model uncertainty. The novel sliding mode control method was proposed to ensure the robustness and accuracy of a controller. LaValle et al. [16] proposed the random trees method to handle the path-planning problem and

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speed up the exploration process. The scientist, Sanyal proposed the efficient and robust control strategy which developed the controller based on the special Euclidean group of rigid body motions, using SE(3) methodology [17], [18]. Xu et al. [19] proposed a cascade ADRC (CADRC) algorithm, which implemented a supplementary high-gain design parameter in the CADRC model. In [20], the adaptive fast terminal sliding mode controller was implemented to track the desired path of multirotor aircraft, which used a nonlinear fast terminal sliding surface to eliminate tracking errors and attenuate fluctuations.

The wind disturbances in the outdoor flight area denote an inevitable issue which has a significant influence on the accuracy of UAV flight control. Many research works have engaged to address this problem. An adaptive control scheme and a robust controller have been implemented in [21], where a synthesized control framework was used to eliminate the wind disturbances by air drag. Song et al. [1] proposed an innovative sliding mode control strategy, where the optimized radial basis function was used to increase the disturbances rejection. Simulation experiments and real-world tests were carried out to corroborate the control performance. In [22], an efficient sliding model control method was proposed, where the adaptive control law was used to eliminate the disturbances. Chen et al. [4] designed an optimized adaptive sliding mode control for multirotor UAVs. The double-layer sliding mode structure was used to estimate the wind disturbances and eliminate tracking errors. Traditional flight control technologies, such as PID control and sliding mode control, have achieved good control effects in practical engineering tasks and can ensure a certain accuracy of trajectory and attitude control. However, in the face of flight control tasks with high real-time requirements and highly dynamic environment, the existing technology still has many defects and challenges caused by a strong reliance on models and the difficulty in the flexible real-time correction of control strategy presets. As flight missions become more and more complicated, the strong nonlinearity of flight environment and the UAV dynamics models have more and more serious impacts on precise control. Compared with the traditional control strategies, artificial-intelligence-based control technology has obvious advantages regarding the accuracy, model dependence, real-time performance and predictability and plays an increasingly important role in the field of UAV wind resistance and high-precision flight control. It has been a strategic technology leading the further development of UAV.

Deep neural network has strong capacities for handling the nonlinear control system [10]. Moon et al. [23] developed a deep reinforcement learning (DRL)-based tracking controller for UAV. The global reward and difference reward functions were deduced for estimating the contribution of each aircraft in the entire system. Thus, the agents were able to receive a high-resolution measurement signal for improving the accuracy of tracking controller. Wu et al. [24] proposed a fixed-wing UAV decision making mechanism which is composed by interfered fluid dynamical system guidance law, back-stepping control loops, and learning-based maneuver controller. The classical control theory, linear ADRC method is implemented to eliminate the external disturbances. The relevant simulations are conducted to test the performance of the proposed method in the obstacle environment. In [25], the data-driven UAV navigation method is proposed. In this framework, the UAV is assumed to fly at the same altitude in the 2-D environment. The A* algorithm is used which generated the training samples under wind disturbances. The DRL model is implemented to learn the navigation solution for UAV. Birnbaum et al. [26] proposed the UAV health monitoring and security system to enhance UAVs' control performance under various external environments. By detecting significant changes, such as wind weather, in the flight, the predefined actions can be activated to address the external disturbances. In [27], the DRL-based landing control method for fixed-wing UAV is proposed. The learning-based model is trained in the wind environment to learn the control policy under external disturbances. Kaifang et al. [28] proposed a maneuver controller for UAV in dynamic unknown environments, using the reinforcement learning solution. The Markov decision process (MDP) combined with the Lyapunov guidance vector field framework is established to describe the flight control model in the target tracking and motion planning flight mission, and certain improvements were achieved in simulation experiments. Kaushik et al. [29] proposed a DRL framework for handling a UAV formation control mission. The innovative reward function was deduced to ensure collision avoidance in formation flight, and a cooperation tracking strategy was used for tracking the target with cooperative information. In [30], a transfer learning-based algorithm was implemented in a DRL-based framework to enable a UAV to use the learning knowledge without retraining. The learning efficiency was verified by the simulation tests.

To solve the above-mentioned problems, this work introduces a novel reinforcement-learning-based UAV tracking control algorithm under wind disturbances. A policy relief (PR) algorithm is proposed to encourage the exploration, and a significance weighting (SW) method is developed for making drones be better adapted to dynamic environments. The random wind disturbance signal has been implemented on numerical dynamic environment in the learning process for improving the robustness of the controller. Furthermore, a novel reward function is designed, which can increase the learning efficiency. In all, this article is developed to study the multirotor aircraft tracking control problem under wind disturbances using reinforcement learning methodology. The contribution includes the following. The main contributions of this study can be summarized as follows.

1) This work provides a new DRL-based UAV tracking control method under wind disturbances. To make the controller be adjustable and robust when there are changes and disturbances in flight area, the PR mechanism is designed to encourage proper exploration in learning phase, which lessens the imbalances between the new knowledge and learning knowledge. Moreover, the SW method is proposed for ensuring the UAV be faster and better adapted to various changes.
2) This study analyzes the characteristics of the whole stage of the tracking flight control task. To achieve optimal performance, a novel reward function is proposed, including sparse reward item to enhance the

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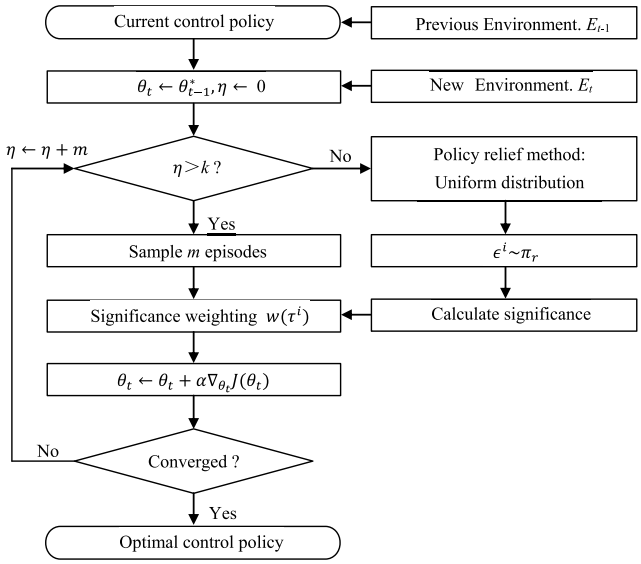

Fig. 1. Flow diagram of the proposed reinforcement-learning-based control algorithm in dynamic environments.

diversity of flight data and increase the efficiency of learning process.

3) Comparative experiments and analysis, including numerical simulation, hardware-in-the-loop (HITL) test, and real-world flight experiments, are conducted to verify the merits of the proposed tracking control algorithm compared with latest relative solutions. The experimental results demonstrate that the proposed algorithm is satisfactory in practical engineering with high accuracy and good robustness.

The rest of this article is organized as follows. Section II describes the proposed method, introducing specific improvements and providing technical details. Section III presents the experimental results and analysis. Finally, Section IV presents conclusions.

II. Incremental Reinforcement Learning Based UAV Control

This section presents the innovative reinforcement-learning-based UAV control algorithm that enables a UAV to learn a control strategy by interaction with a dynamic environment. The problem of UAV tracking control is formulated as an MDP problem, and a novel deep deterministic policy gradient (DDPG) algorithm is used to construct a UAV tracking control model. The state space, action space, and reward function are presented. The incremental-learning-based mechanism is implemented into an RL-based controller to make a UAV adjustable to a dynamic environment. To enhance the robustness and stability of a UAV under unknown disturbances, the DDPG tracking control framework is optimized by PR and SW. Fig. 1 illustrates the process of the proposed method.

A. Problem Description

In this study, an innovative DDPG-based control algorithm has been used to solve the MDP problems with continuous actions. By representing the policy as a policy function, a deterministic policy gradient method can map a state to a deterministic action. In this study, external disturbances are regarded as changes in a flight environment, and changes in environments have been formulated as a series of split tasks in time series where each task correlates with a specific characteristic in the correlated time series. The state transition function has drift during changes in an environment, but the state and action spaces remain the same. Assume that a dynamic environment set $\mathcal{D}$ is $\mathcal{D} = [E_1, \ldots, E_{t-1}, E_t, \ldots]$, where E_t represents a specific characteristic of a dynamic environment at the tth time frames. The main purpose of an RL-based model is to generate an optimal control strategy that can maximize the desired persistent reward $J(\theta)$ under parameter θ. The desired long-term reward is expressed as follows:

$$J(\theta) = \mathbb{E}_{\epsilon \sim \pi(\epsilon)}[r(\epsilon)] = \mathbb{E}_{\epsilon \sim \pi(\epsilon)}\left[\sum_{i=0}^{\infty} \gamma^i r_i\right] \tag{1}$$

where $\epsilon = \{s_0, a_0, s_1, a_1, \ldots\}$ means a training episode, r_i is the immediate reward after interaction with an environment, and γ is the discount factor. According to the theorem of policy gradient, the formula for the basic policy gradient algorithm is defined by the following equation:

$$\begin{aligned}\nabla_\theta J(\pi_\theta) &= \int_\epsilon \nabla_\theta \pi_\theta(s) \nabla_a Q^\pi(s,a)\Big|_{a=\pi_\theta(s)} \mathrm{d}\epsilon \\ &= \mathbb{E}_{s\sim\pi_\epsilon}\left[\nabla_\theta \pi_\theta(s) \nabla_a Q^\pi(s,a)|_{a=\pi_\theta(s)}\right].\end{aligned} \tag{2}$$

During the time of the learning process, the policy approximation is implemented to optimize the parameters of model, which can be expressed as follows:

$$\theta^*_{t-1} = \arg\max_{\theta\in\mathbb{R}^d} J_{E_{t-1}}(\pi_\theta). \tag{3}$$

The external disturbances are described as changes in environmental characteristic. The function of the designed incremental learning method is to enable a UAV to update its parameters θ^*_{t-1} to an optimized value θ^*_t automatically when the environment characteristics transfer into E_t at the tth time

$$\theta^*_t = \arg\max_{\theta\in\mathbb{R}^d} J_{E_t}(\pi_\theta). \tag{4}$$

In this way, an agent is able to update its existing knowledge according to the changes in a dynamic environment. An innovative reinforcement learning control method is integrated with the incremental learning process for the UAV control, which includes PR and an SW mechanism. The detailed implementations are demonstrated in Section II-C and Section II-D.

B. Design of System State and Reward Function

A reinforcement-learning-based control strategy, which enables a UAV to track the target accurately, is developed. In the flight mission, the movement relationship between a UAV and a target plays an imperative part in the control strategy design. The 10-D vector S^*_t is used to represent the state of the current UAV which is chasing the target in a flight task, and it is given by the following equation:

$$S^*_t = \{\Delta P_t, \Delta\Phi_t, \Delta V_t, B_t\}^T \in \mathbf{R}^{10}. \tag{5}$$

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S_t^* could be divided into several subset vectors. $\Delta P_t = (\Delta x, \Delta y, \Delta z)$ means the relative distance between the current vehicle and the target in three axes. $\Delta\Phi_c = (\Delta\phi, \Delta\theta, \Delta\psi)$ represents the relative attitude angle, including roll, pitch, and yaw. The relative velocity between the UAV and the target is denoted as $V_c = (v_x, v_y, v_z)$, and B_c is the battery state of the vehicle. Considering the kinematic relationship between the UAV and the target in tracking flight, as well as the past flight state, we define the tracking control state as follows:

$$S_t = \left\{S_t^*, S_{t-1}^*, S_{t-2}^*, S_{t-3}^*\right\}^T \in \mathbf{R}^{40}. \tag{6}$$

The state space in the designed tracking control model contains the current flight state and past three history state data frames. A multirotor aircraft is mainly controlled by the pulling force of the rotors. By changing the rotation speed of the front and rear rotors, the rotation speed difference is generated, and the pitch control is realized through the differential. Therefore, the action space could be expressed as $a = \{u_1, u_2, u_3, u_4, u_5, u_6\}$, where u_i means the rotating speed of a rotor i.

The proper design of the reward function is one of the most crucial factors in the RL-based model construction process. An effective reward function can provide a faster convergence and a better performance in experiments. This study analyzes the characteristics of the whole stage of the tracking flight control task. To reach the optimal performance, a novel reward function, including state deviation value, energy consumption, and sparse reward obtained when a vehicle is reaching the target interval, is designed as follows:

$$\begin{aligned} r_t = &-(w_1|\Delta P_t| + w_2|\Delta\Phi_t| + w_3|\Delta V_t| \\ &+ w_4|B_t - B_{t-1}|) + r_P^+ + r_V^+ \end{aligned} \tag{7}$$

where w_i denotes the weight of the corresponding terms. Immediate reward implements deviation value as a penalty item, including position deviation, velocity deviation, and attitude angle deviation. When the control performance is not adequate, the penalty item will be large, and the penalty value will approach zero when an aircraft approaches the steady state. Setting an immediate reward in this form demonstrates that the agent's policy goal is to control a UAV to fly stably at a steady-state point throughout the flight mission. r_P^+, r_V^+, and r_{done}^- denote the sparse reward items improving the sample diversity of flight data and enhancing the learning efficiency of the agent.

C. PR Method

In a dynamic changing environment, the reinforcement-learning-based control model tends to use a local optimum instead of the whole state–action space, which may lead the model to fall into a local optimum [31]. The reason for this is that an agent has insufficient exploration during in the learning process, and its parameters are overfitted to the existing environment. To make an agent be adjustable to the environmental changes and strengthen an appropriate exploration, this study designs a PR method. In particular, θ represents the current parameter set of an action network in an environment E_t, and π_θ is the policy derived from the current parameters. Thus, the PR strategy is designed as follows:

$$\pi_r(a \mid s) = \begin{cases} \text{Un}(A(s)), & \eta \le k \\ \pi_\theta(a \mid s), & \eta > k \end{cases} \tag{8}$$

where $\eta = \eta_0 * 0.999^\lambda$, representing the PR probability, where η_0 is the initial probability of the PR mechanism, and λ is the number of training steps; k is the presetting probability threshold. When $\eta > k$, the action is generated by the current policy π_θ; otherwise, the action is calculated by the uniform distribution $\text{Un}(A(s))$, where $A(s)$ is the available action set corresponding to the state space.

To make the PR mechanism practicable and unbiased, the general statistic technique [32] and importance sampling [33] are implemented to calculate the policy gradient. Think of the object of reinforcement-learning-based model in (1), and $r(\epsilon) = \sum_{i=0}^{\infty} \gamma^i r_i$ is the entire reward of an episode ϵ. In the PR mechanism, it is in necessary to evaluate the expected value of $r(\epsilon)$ under the initial distribution of $\epsilon \sim \pi_\theta(\epsilon)$, while it is going to sample from a novel policy distribution $\epsilon \sim \pi_r(\epsilon)$ in the dynamic changing environment. Therefore, the inspection of importance sampling is given as follows:

$$\begin{aligned} \mathbb{E}_{\epsilon\sim\pi_\theta(\epsilon)}[r(\epsilon)] &= \int_\epsilon \pi_\theta(\epsilon) r(\epsilon) \mathrm{d}\epsilon \\ &= \int_\epsilon \frac{\pi_\theta(\epsilon)}{\pi_r(\epsilon)} r(\epsilon) \pi_r(\epsilon) \mathrm{d}\epsilon \\ &= \mathbb{E}_{\epsilon\sim\pi_r(\epsilon)}\left[\frac{\pi_\theta(\epsilon)}{\pi_r(\epsilon)} r(\epsilon)\right]. \end{aligned} \tag{9}$$

Furthermore, the calculation process of the policy gradient is rewritten as follows:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\epsilon\sim\pi_r(\epsilon)}\left[\frac{\pi_\theta(\epsilon)}{\pi_r(\epsilon)} \nabla_\theta \pi_\theta(\epsilon) \nabla_a Q^{\pi_\theta}(\epsilon)\right]. \tag{10}$$

The design of the PR mechanism is inspired by the conventional RL technique named the epsilon-greedy exploration. The action policy is implemented to incorporate certain types of random actions that can enhance the exploration of an agent. This study is inspired by the epsilon-greedy algorithm and purposes an appropriate exploration strategy for the implementation of THE policy gradient method. To achieve a tradeoff between exploration and exploitation, a significance sampling mechanism is developed to achieve collaboration with the PR method.

D. SW Method

The control strategy obtained from the existing environments may provide a modest performance in a dynamic changing environment since its parameters have learned the knowledge about characteristics from the original environment, and the policy may not be optimal for new environment. However, direct application of parameters may decrease the performance of an agent in the long term when interacting with a new environment because initializing the neural networks parameters may cause them to fall into a local optimal point which is overfitted to the previous environments [34], [35], [36]. In addition, parameters may transfer from current local optimal point to another, because the optimization can be

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influenced by pathological curves. In the learning process, the model tends to perform the action that is feasible for the existing environment. This is because an agent has to be trained to have the prior knowledge necessary for handling the problems in the previous environment. Therefore, it is essential to make an agent be adjustable to the environmental changes. In the dynamic environment, the example episodes may not achieve the optimal performance because their parameters are updated based on information from the previous environment. Different episodes may obtain various rewards under a dynamic environment. The training episodes with more rewards are supposed to have more importance and more information in the learning process [35], [37]. Therefore, in this study, a higher weight is assigned to episodes with a higher reward value, which is beneficial for improving the convergence speed of the model. Inspired by the mentioned perception and related literature, the representation of designed significance weight is given by the following equation:

$$w(\epsilon^i) = \frac{1}{\rho}\left(r(\epsilon^i) + \xi\right), \quad i = 1, \ldots, n \tag{11}$$

where the ξ is a tiny mathematical constant, n is the number of samples in a mini-batch, and ρ is the normalization item making the weight metric of mini-batch equivalent to one. Based on the DDPG theorem [38], the direct gradient ascent function in (2) can be rewritten as follows:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\epsilon\sim\pi_r(\epsilon)}\left[w(\epsilon)\nabla_\theta \pi_\theta(\epsilon)\nabla_a Q^{\pi_\theta}(\epsilon)\right]. \tag{12}$$

Based on the above SW method, an agent can update its parameters tending to the region of parameter space that is more appropriate for the current dynamic environment.

E. Integrated Method

Algorithm 1 demonstrates how the designed incremental reinforcement learning method works for UAV control in dynamic environments. Wind disturbance is regarded as changes in the environment, and the controller is designed so that it can adjust its characteristic in a dynamic environment using the above-mentioned method. A random wind disturbance signal has been implemented on numerical dynamic environment in the learning process. The model is designed to be able to interact with various environments represented as a set $\mathcal{D} = [E_1, \ldots, E_{t-1}, E_t, \ldots]$, where E_t denotes a type of wind disturbance characteristic at the tth time series. In the first interaction with the environment, an agent learns the original policy through a standard policy gradient strategy, as presented in lines 2 and 3. During the following interaction phase, the neural network parameters are initialized based on the learned policy, which is optimal for the preceding environment, as shown in line 5. In this way, the agent is able to directly use the information from the previous environment since it has learned the basic feature representation of the environment and state–action spaces. Then, the PR algorithm is implemented to achieve an appropriate exploration with higher diversity, as given in line 6. To improve the convergence speed of the learning process, the SW method is adopted to assign higher significance weights to the samples accomplishing with higher total rewards, as demonstrated in lines 7 and 9.

Algorithm 1 Incremental Reinforcement-Learning-Based UAV Control Method in Dynamic Environment

Input: Environment serials$\mathcal{D} = [E_1, \ldots, E_{t-1}, E_t, \ldots]$; current time step t(>0); parameter set;
Output: Optimal policy parameters
Data: Testing set x

1 **if** *t equals to 1* **then**
2 Initialize the parameters from beginning
while *no converged* **do**
3 Sampled m episodes:
$\nabla_\theta J(\theta) = \mathbb{E}_{\epsilon\sim\pi_r(\epsilon)}[w(\epsilon)\nabla_\theta\pi_\theta(\epsilon)\nabla_a Q^{\pi_\theta}(\epsilon)]$
$\boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_t + \alpha\nabla_{\boldsymbol{\theta}_t} J(\boldsymbol{\theta}_t)$
4 **else**
5 Update the training episodes $\eta \leftarrow 0$
Update the parameter set
while *no converged* **do**
6 **if** $\eta \le k$ **then**
7 $\pi_r(a \mid s) = \text{Uniform}\ (A(s)), \forall s$
Select m samples from π_r
Calculate the significance weights for learning episodes by (11)
$\nabla_\theta J(\theta) = \mathbb{E}_{\epsilon\sim\pi_r(\epsilon)}[w(\epsilon)\frac{\pi_\theta(\epsilon)}{\pi_r(\epsilon)}\nabla_\theta\pi_\theta(\epsilon)\nabla_a Q^{\pi_\theta}(\epsilon)]$
8 **else**
9 Select m samples from π_θ
Calculate the significance weights for learning episodes by (11)
$\nabla_\theta J(\theta) = \mathbb{E}_{\epsilon\sim\pi_r(\epsilon)}[w(\epsilon)\nabla_\theta\pi_\theta(\epsilon)\nabla_a Q^{\pi_\theta}(\epsilon)]$
10 $\boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_t + \alpha\nabla_{\boldsymbol{\theta}_t} J(\boldsymbol{\theta}_t)$
$\eta \leftarrow \eta + m$
$u \leftarrow u + \Delta u$

Furthermore, the policy networks' parameters are updated in proportion to achieve the gradient till convergence, as shown in line 10. Finally, the policy suitable for dynamic changing environment is obtained.

III. Experimental Results

To evaluate the performance of the proposed incremental reinforcement-learning-based UAV control strategy in dynamic changing environments, a series of experiments were performed, including numerical simulation test, multiple wind disturbance tests, HITL experiments, and a real-world flight test. In addition, to investigate the designed reinforcement-learning-based control algorithm further, it was compared with several state-of-the-art techniques, including the RL-based method and classic control method. Two evaluation criteria—root mean square error (RMSE) indicators and maximum (MAX) indicators of tracking errors—are used to quantitatively assess the tracking control performance of the proposed method. M is defined as the samples size, and e_i denotes the position tracking error. The corresponding formulations are given as follows:

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{i=1}^{M} e_i^2} \quad \forall i \in A \tag{13}$$

$$\text{MAX} = \max(|e(i)|). \tag{14}$$

The experimental setup and results' analysis are described in the following.

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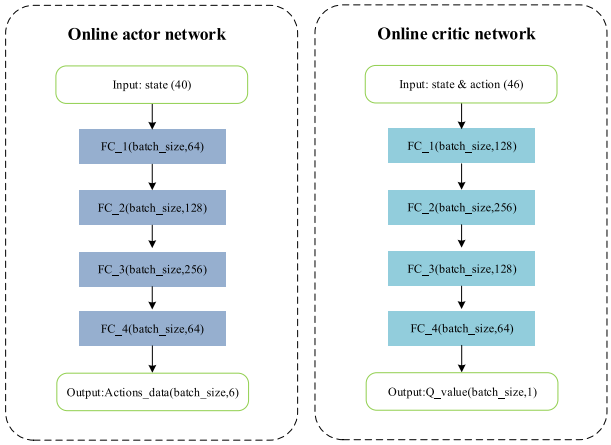

Fig. 2. Framework of the proposed method.

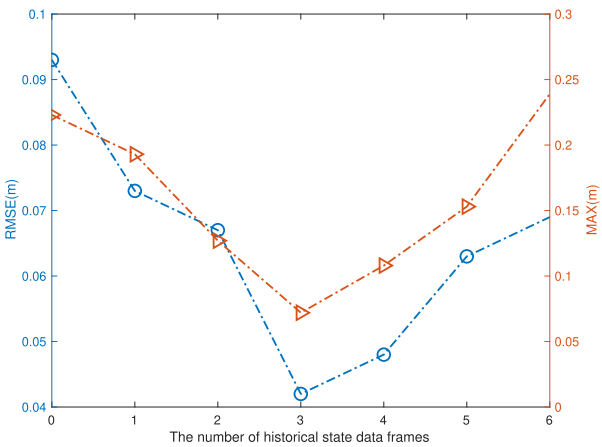

Fig. 3. Test accuracy of the proposed method with increasing number of historical state data frames.

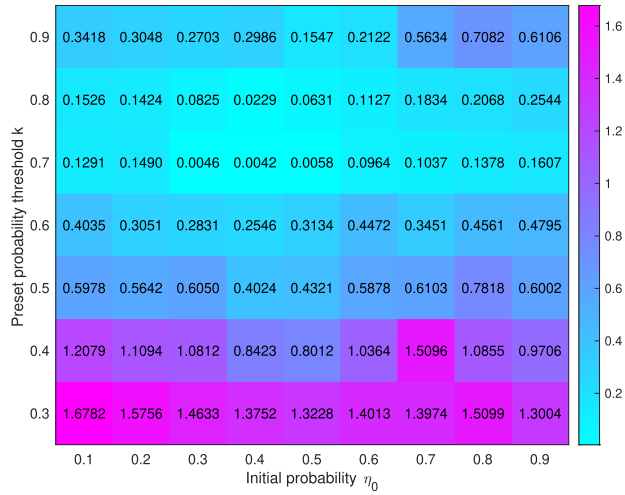

Fig. 4. Relationship between control accuracy and parameter set (η_0, k) in numerical test.

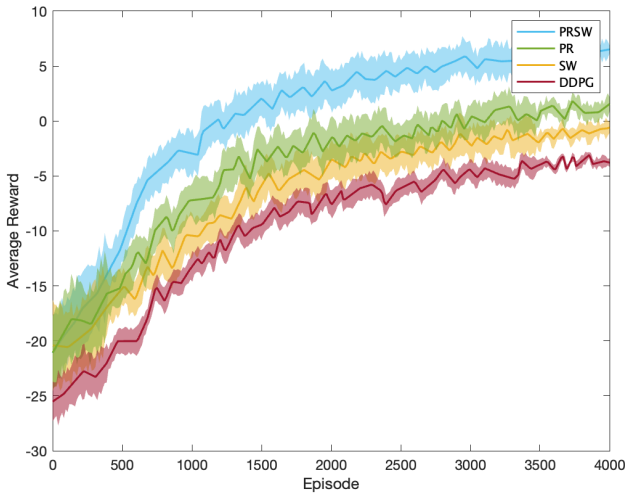

Fig. 5. Average rewards per episode.

A. Experimental Configuration and Parameters Setting

The proposed reinforcement-learning-based control algorithm was implemented in Python using TensorFlow [39]. The DDPG framework, consisting of online critic network, target critic network, online actor network, and target actor network, was used. The network architectures of the target critic network and target actor network were the same as those of the online critic network and online actor network, respectively. The network structure of the designed method is shown in Fig. 2, where it can be seen that there were four hidden layers in both the actor network and critic network. The input and output of the actor network were a system state $s \in \mathbf{R}^{40}$ and action $a \in \mathbf{R}^{6}$, respectively. The number of neural points in the actor network's hidden layers was 64, 128, 256, and 64, respectively. The architecture of the designed incremental reinforcement learning control model is shown in Fig. 2. For the critic network, the hidden layer structure was 128–256–128–64. There were 4000 episodes in the training phase for learning.

Since there is no sufficient mature mathematical theory to support the process of setting parameters, it is not easy to choosing appropriate parameters' value in the RL-based model, which requires extensive trial and error and practical experience [40], [41]. In the proposed method, the tracking control state contains the current flight state and the past three flight data frames. That is, because historical flight data usually have strong correlations to the current flight state, which are helpful for neural network to obtain better comprehensive estimation for current state and generate more appropriate control signal. In experiments, we noted that the control accuracy in terms of RMSE and maximum errors could be improved by initially increasing the number of past data frames in flight state, which is shown in Fig. 3. It is observable that the control errors decrease rapidly and then tend to improve with the growing number of historical data frames. Thus, we have chosen the past three historical data as the part of current flight state to obtain optimum performance.

For the weight coefficient of each item in the reward function, we empirically select the value of parameters in reward function. All the tracking control errors are defined as negative penalty item, which will have large value when the control results are neither stable nor accurate. When the policy controls vehicle approach steady state, the penalty value is close to zero. Setting an immediate reward in this way aims to control UAV to stably fly at a desired state throughout

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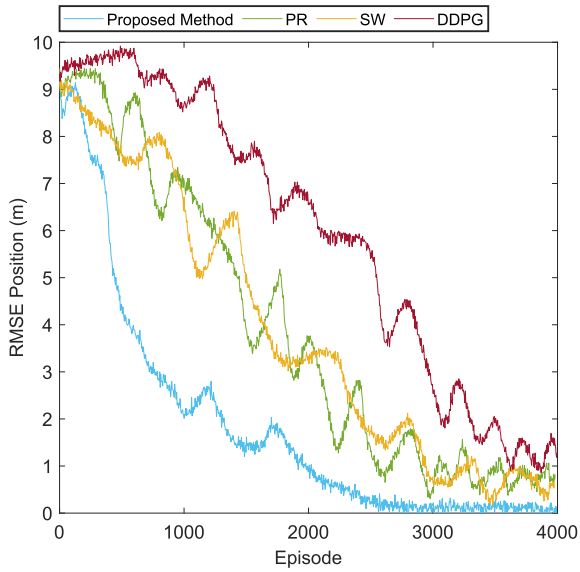

Fig. 6. RMSE performance per episode.

the flight mission. Specifically, we set $w_1 = 0.4$, $w_2 = 0.4$, $w_3 = 0.2$, and $w_4 = 0.02$. That is, because the position control and attitude control have more importance in the designed control method, and the velocity control item and energy cost item have been assigned with less weight for their low priority comparing with the first two items. Spare reward is activated when the UAV reached the specific flight state, such as tracking with high accuracy, reaching unavailable state. r_P^+ is the position precision reward item. The agent will obtain a sparse positive reward once the position error is within 0.2 m. The value of r_P^+ is 5 if these conditions are met; otherwise, r_P^+ is zero. r_V^+ is the velocity precision reward item, and it will be activated with the positive value 5 once the velocity tracking error is within 0.1 m/s. When the control command guides the drone to reach the unavailable state, the sparse plenty r_{done}^- will be implemented with the negative reward -100. The mentioned three sparse rewards will only be activated under some certain conditions. If the specific condition is met, the reward will be triggered with specific rewards. Otherwise, it will be set to zero.

In the proposed PR model, the setting of preset probability threshold and initial probability might tend to influence the control results. The cross-validation method is used to select the optimal parameter set (η_0, k), in which the candidates' set of η_0 is chosen as $[0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]$, and the candidates' set of k is $[0.4, 0.5, 0.6, 0.7, 0.8, 0.9]$. Fig. 4 illustrates the relationship between RMSE value and parameters' setting (η_0, k). It can be seen that the optimal performance can be achieved with the parameter set $(\eta_0 = 0.4, k = 0.7)$.

B. Training in Numerical Simulation Environments

The random initial flight state selection strategy and wind disturbance were implemented into the numerical system to simulate changes in a dynamic environment. The ablation experiments were conducted to explore how do PR and SW strategies influence the results of the designed control model. Four methods were used for learning in dynamic changing environments as follows.

1) *Proposed Method*: This method used both the PR (PR) method and the SW method in experiments, and it was denoted as the PRSW control method;

2) *PR*: This method implemented the PR strategy without using the SW strategy in testing;

3) *SW*: This method implemented the SW strategy without the PR strategy in testing;

4) *DDPG*: In this method, neither the PR nor SW mechanisms were used in the DDPG framework, which was used as a baseline in the ablation study.

The average reward over a batch of learning episodes was used to evaluate and analyze the performance of the reinforcement-learning-based tracking control method. The average reward was defined by: $G_{\text{Ave}} = (1/N_e N_t) \sum_{i=1}^{N_e} \sum_{t=1}^{N_t} r_t^i$, where G_{Ave} represents the average return through bunch of N_e episodes, and r_t^i means the received reward for accomplishing the learned control action at time step t of certain episodes e. In the learning stage, the average reward was evaluated every ten episodes.

Fig. 5 illustrates the learning results in terms of G_{Ave} of different variations in the methods in the ablation study. First, the results of the baseline methods DDPG and SW were compared, and it was concluded that the SW mechanism could improve the performance of the baseline algorithm. In the learning phases, the SW strategy was used to expand the opportunities of accepting samples with high importance, which enabled improving the capability in terms of average reward. The SW strategy could encourage an appropriate exploration in the initial learning stage and accelerate the convergence speed. Next, the performance in terms of G_{Ave} of the PR mechanism was also more capable than the variation in baseline, which demonstrates that the PR strategy could effectively improve the controller's adaption ability to dynamic changing environments. Besides, the average reward of PRSW, which included the PR and SW mechanisms, increased the fastest and accomplished the best results, among all the methods, almost through the entire learning phase. In addition, G_{Ave} of the PRSW grew rapidly and converged faster than those of the other three variants of methods. The ablation experiments in the numerical simulation environment have shown that the proposed method can significantly improve the learning effectiveness.

The evaluation metric of RMSE is also provided in this section. First, from the learning process represented by the RMSE position performance in Fig. 6, the proposed method learns faster and achieves the better convergence speed than other methods. In the beginning of the learning process, the proposed method dose not achieve the accuracy control performance and the RMSE indicators in the early episodes are higher. That is because the parameters of neural networks are not converged to the optimal status in the initial stage. The proposed PR and SW algorithm are able to provide sufficient exploration and guidance for the learning process, which contributes to improving the stability and quality of the training process. Moreover, the detailed convergence speed, computation energy, RMSE indicator, and inference time are

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TABLE I
EVALUATION METRIC ABOUT LEARNING PROCESS

Method	Inference time (ms)	RMSE (m)	Convergence Speed (episodes)	Computation consumption (MB)
Proposed method	0.71	0.0141	3870	3.8
PR	1.43	0.7688	4458	5.4
SW	1.82	0.7930	4659	5.5
DDPG	3.49	1.6192	5207	5.1

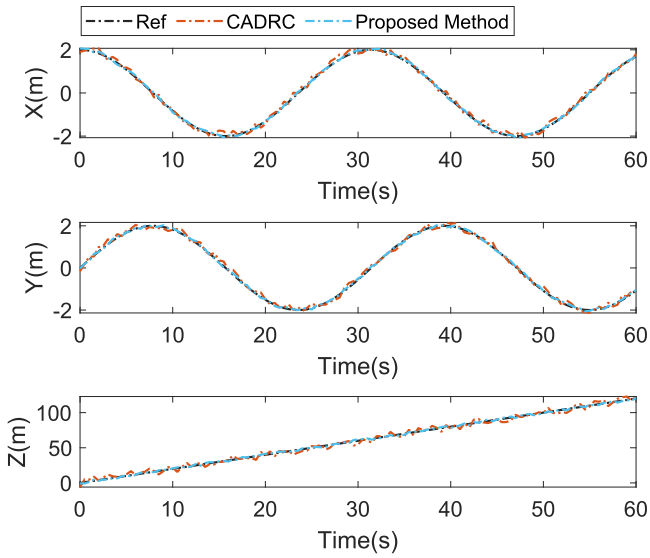

Fig. 7. UAV's desired position and actual position.

listed on the Table I. As demonstrated in Table I, the inference time of the proposed method is faster than other solutions while using the similar computation consumption. It can be seen from the above analysis that the learning process of the proposed method achieves the stability and convergence within 4000 episodes while other three methods still suffer from lower rewards and higher errors in the limited training time. According to the above analysis, the stability and effectiveness about the learning process of the proposed method are verified.

The tracking control experiments were conducted in numerical simulation for digging into the control performance of the proposed method and to compare it with the performance of the state-of-the-art control method, the CADRC algorithm. The vehicle started from the initial position (2, 0, 0) and began to track the desired trajectory of movements. The comparison of the trajectory tracking results and attitude control errors of the proposed method and the CADRC method in three channels is shown in Figs. 7 and 8, while the position tracking errors and attitude control errors are shown in Fig. 9. The solid black line represents the desired tracking signal, the blue dashed line represents the response tracking result of the PRSW, and the orange dashed line denotes the response tracking result of classical control-based CADRC method. When the UAV was tracking the desired trajectory, the position tracking errors of the proposed RL-based control method were within ± 0.15 m, and the attitude control errors were no more than $\pm 0.2°$. In practical application, we explain the mean position tracking error within 0.5 m as the acceptable performance, because the UAV is able to conduct lots of flight missions

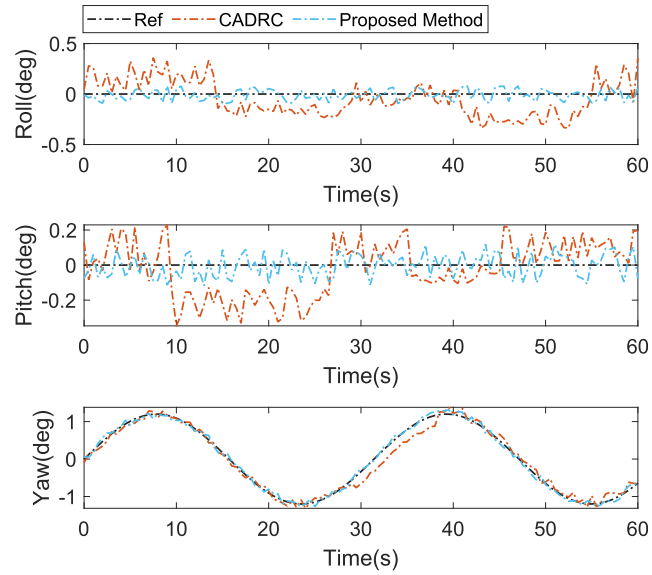

Fig. 8. Attitude control results in numerical experiments.

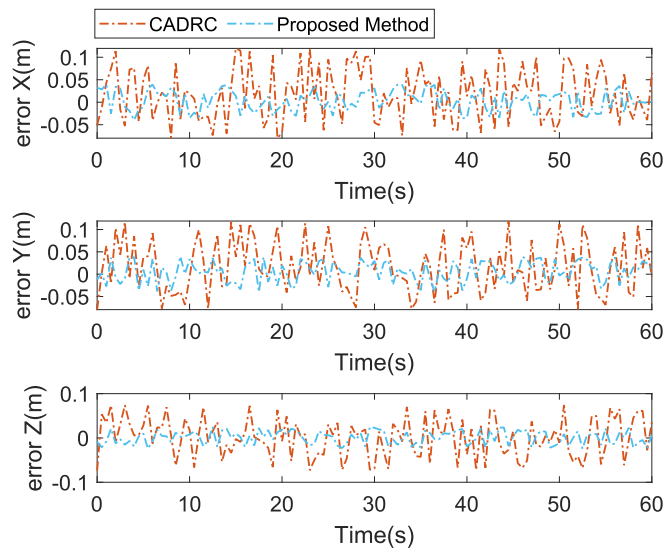

Fig. 9. Position control errors in numerical experiments.

under that control precision. The corresponding results of numerical errors are summarized in Table II. In Table II, it can be seen that the proposed method achieved better control results than CADRC; however, the gap is not very large in the first set of numerical simulation. The reason is that in the ideal environment of numerical experiment, it is easy for many SOTA controllers to achieve good performance without considering any real-world restrictions [1], [4]. According to the results, compared with the vehicle implemented with the CADRC algorithm, the UAV implemented with the proposed RL-based control algorithm had a better ability to track the desired trajectory movements and achieved smaller control errors in terms of position, velocity, acceleration, and attitude control. The numerical experimental results have demonstrated that the designed method can be effectively applied to the UAV tracking control flight in the simulation environments.

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TABLE II
TRACKING ERRORS IN NUMERICAL SIMULATION WITHOUT EXTERNAL DISTURBANCES

Method	Pos RMSE (m)	Vel RMSE (m/s)	Acc RMSE (m/s^2)	Att RMSE (deg)
Proposed method	**0.0141**	**0.0174**	**0.0220**	**0.0515**
CADRC	0.0755	0.0879	0.1335	0.3320

C. HITL Experiments Under Wind Disturbances

For the purpose of testing the robustness and feasibility of the designed control algorithm under wind disturbances, the HITL experimental platform has been established. The HITL platform, as shown in Fig. 10(a), contains flight computer for implementing control model, environment simulator, actuators, batteries, and ground station. The flight control computer is the core part and determines the performance and safety of the system. The proposed RL-based control model of UAV is running on the flight computer where the control model can receive the feedback of multirotor's flight state (i.e., velocity, position, and attitude angle) and calculate the control command to actuators. The flight computer receives the remote commands of the ground console through the serial port utility and executes the corresponding tasks. The sampling frequency of the HITS system is 600 Hz, which meets the requirements of high-fidelity simulation system. We adopt Airsim [42] as flight environment simulator, including environment module, virtual sensor modules, airframes model, physical engine, rendering interface, and an interface API for the control model. In the simulation test, Airsim consumes the actuator signals generated by the proposed controller to figure out force and thrust generated by each actuator. This is then used by the physics engine to compute the kinetic properties of the drone. This in turn generates simulated sensor data and feed it back to the flight controller. The virtual sensor model of IMU, gyroscope, magnetometer (compass), and barometer implemented in HITL is MPU6500, GY9250, bmm150, and icp10111, respectively. The version of this simulator is AirsimV1.8 with UE5. The HILT experimental platform is applied on a computer with Intel core i7-8700 CPU, a NVIDIA GTX-2080 GPU, and 16-GB RAM. After receiving the remote control instructions from the console software, the flight control computer generates control instructions according to flight states and finally drives the motor and steering gear to make the UAV take automatic flight control. The flight control computer receives the remote control commands of the ground console through the serial port server and executes the corresponding flight tasks. The flight control computer publishes the state data and actuator data of the aircraft to the 2-D and 3-D situation software through the serial port server. Finally, the flight information of the entire flying stage was transferred to the ground state for visualization and analysis. Fig. 10(b) shows the experimental platform consists of a self-developed UAV. Weighing approximately 5 kg with a maximum takeoff weight of about 11 kg, the self-developed UAV can easily carry equipment as heavy as a Lidar VLP-16. Used with a 32 000-mAh battery, it can fly for up to 45 min. Its diagonal wheelbase is 1045 mm, and the maximum flight speed is 9 m/s.

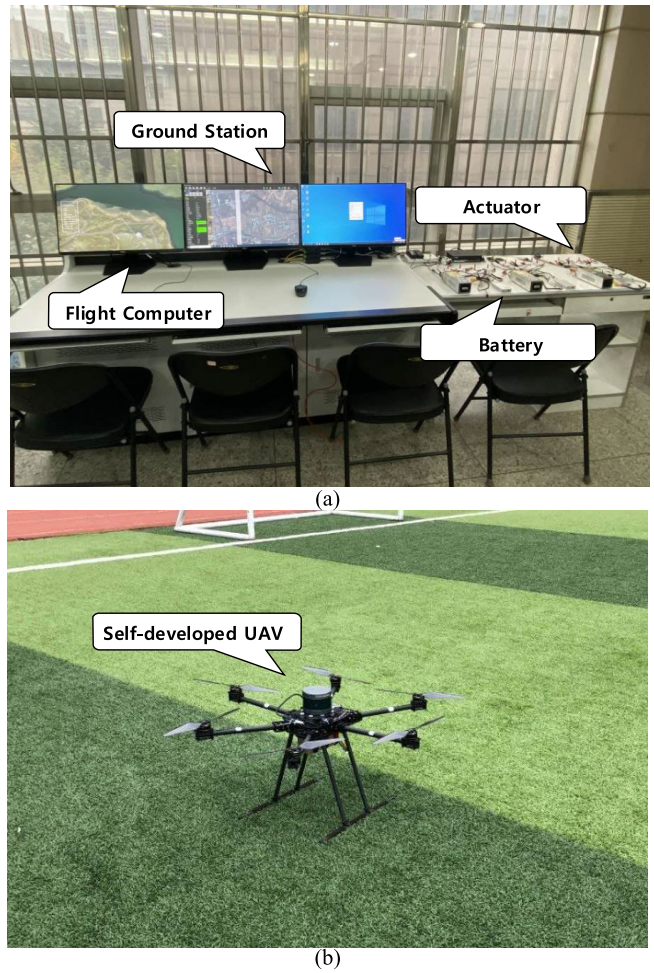

Fig. 10. HITL simulation platform and self-developed UAV. (a) HITL simulation platform. (b) Self-developed UAV.

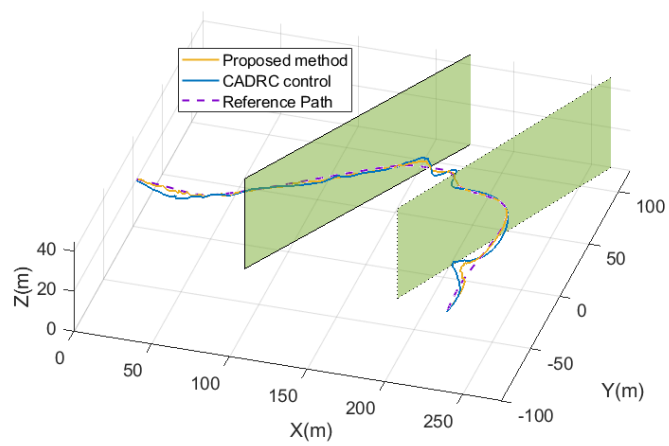

Fig. 11. Trajectory tracking control results in various wind environments.

Since the Airsim simulator had a strong capacity for simulating the wind field environment in a flying mission, we applied three different types of wind disturbances using the environment simulator for demonstrating the disturbance rejection performance of the proposed method. As shown in Fig. 11, in the first zone of the flight field, wind disturbances were constant; in the middle zone of the flight region, wind disturbances were set to the changing wind, which meant the

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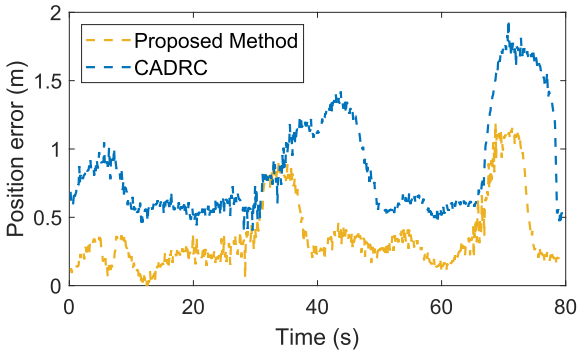

Fig. 12. Position control errors in various wind environments.

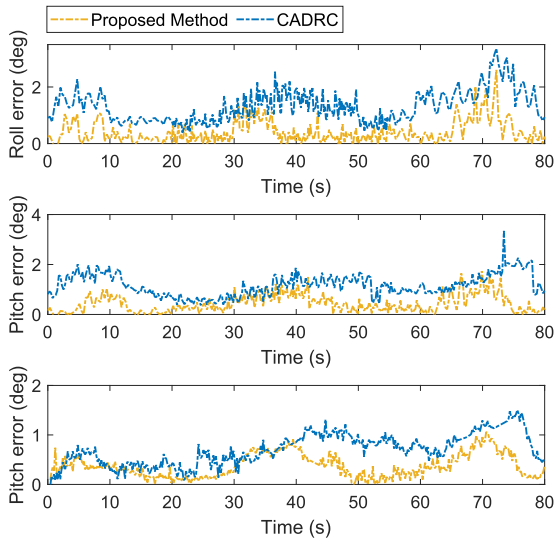

Fig. 13. Attitude control errors in various wind environments.

speed and torque of wind were randomly generated in the simulator; finally, in the last zone, there was a strong wind gust acting on the last segment of the flight trajectory. As shown in Fig. 12, both the PRSW and CADRC control algorithms could fly along the given trajectory in the whole wind field and the deviation of the RL-based method was smaller than that of the CADRC. The attitude control errors while the vehicle was passing through the wind region are shown in Fig. 13, where it can be seen that the stabilization performance of the proposed algorithm was less affected by the wind disturbance type compared with those of the other control methods. Specifically, the angle offset of the UAV using the proposed method had a slight shift at the beginning of interference and then returned to the stable state in the middle and late periods of wind disturbance, which made it faster than another method. Thus, the proposed control algorithm had good robustness under various types of wind disturbances.

The comparison experimental results under wind disturbances are analyzed and discussed to test the performance of the proposed tracking control method comparing with other SOTA methods further. The experimental setting of wind disturbances is the same as those in the above-mentioned experiments. As shown in Table III, the RMSE performance of the proposed method in multievaluation channels is smallest

TABLE III

COMPARISON EXPERIMENTS WITH LATEST METHODS UNDER WIND DISTURBANCES

Channel	Evaluation Metric	**Proposed method**	Method [19]	Method [4]	Method [23]	Method [29]
X(m)	RMSE	**0.2641**	0.8243	0.5942	0.3574	0.7029
	MAX	0.8960	2.0665	0.9471	**0.8223**	2.5113
Y(m)	RMSE	**0.1907**	0.6877	0.6418	0.4956	0.6581
	MAX	**0.7352**	1.8024	1.2506	1.2838	1.3725
Z(m)	RMSE	**0.2382**	0.7535	0.6523	0.4014	0.8618
	MAX	**0.6608**	1.8658	0.8302	0.9954	3.7009
Vel(m/s)	RMSE	**0.2596**	0.8134	0.7469	0.6854	0.7157
	MAX	**0.5843**	2.2324	1.9576	1.3003	1.4922
Acc(m/s^2)	RMSE	**0.2740**	0.8975	0.6706	0.7419	0.8595
	MAX	**0.4501**	1.9785	1.9572	1.2218	2.2467
Att(deg)	RMSE	**0.8766**	6.6235	5.8491	6.2340	7.6788
	MAX	**2.4501**	10.9785	9.9572	8.2218	9.2467

TABLE IV

RMSE PERFORMANCE OF THE PROPOSED METHOD AND CADRC CONTROLLER IN THE REAL-WORLD ENVIRONMENT UNDER WIND DISTURBANCES

Method	Pos(m)	Vel(m/s)	Acc(m/s^2)	Att (deg)
Proposed method	**0.2257**	**0.2429**	**0.2691**	**1.4359**
CADRC	1.6356	1.8283	2.2177	4.6020

among other four latest controllers. Moreover, the maximum errors of the proposed method are also significantly smaller than other methods. That is because the designed RL-based solution has learned appropriate control policy under external disturbances in the learning phase and provides robustness capability to disturbance force. The comparison results show that the proposed method achieves better robustness compared with latest methods.

D. Real-World Flight Experiments

To investigate the reliability of the proposed control algorithm further, it was analyzed experimentally using a real-world multirotor UAV platform. Fig. 10(b) shows the designed self-developed multirotor UAV, which uses the NVIDIA Jetson Nano as the flight control computer. The AH100B sensor was adopted as the inertial measurement unit (IMU) model, which could precisely measure the flight state parameters (i.e., attitude angle, angle rate, and velocity). A Lidar system was implemented to assist in the measurement of real-time data. The flight data were collected and transferred to the ground state for observation. Fig. 14 shows the real-world flight test setup. The multirotor aircraft was commanded to track the desired path in experiments. Comparative experiments on the proposed method and CADRC controller were implemented under the same conditions.

As shown in Fig. 15, the multirotor using the control method could track the desired path with high precision. The initial position of the UAV was defined as (10, 5, 0). The UAV took off from the initial position, rising up to 30 m, and then started to move along the elliptical path with an average speed of approximately 2.8 m/s. After flying around the elliptical path, the vehicle returned and began landing in the initial position. The curve of airspeed changes is shown in Fig. 16, where it can be seen that the airspeed of the UAV fluctuated during

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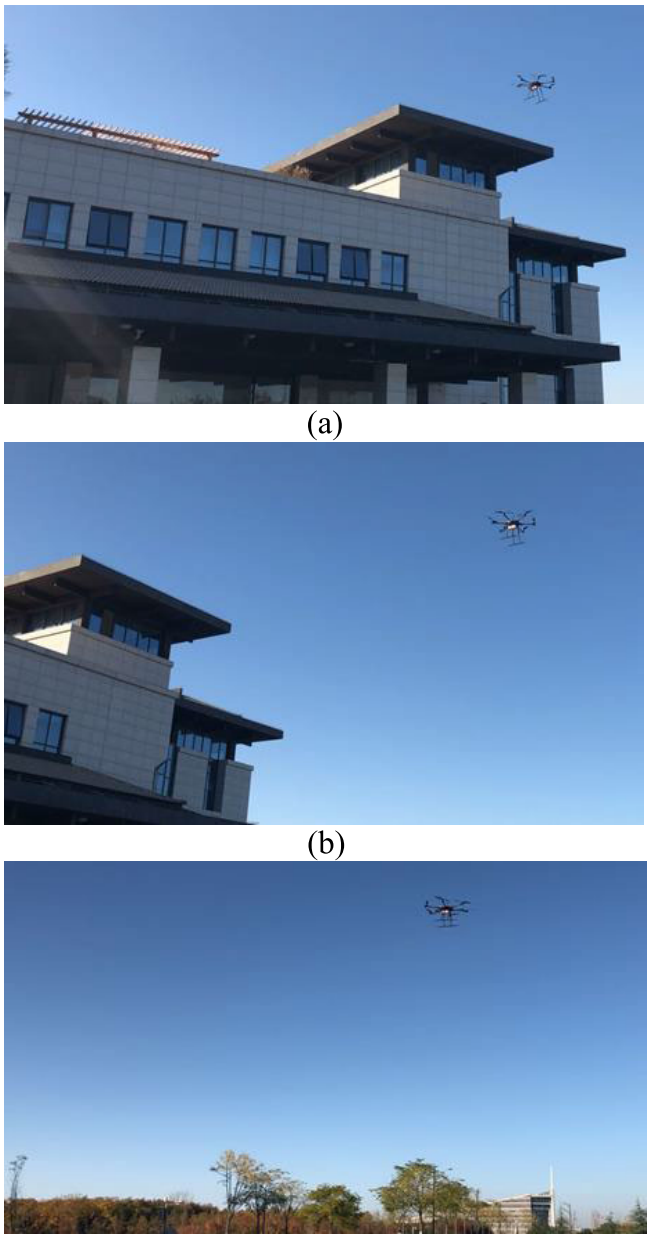

Fig. 14. UAV tracking scene in real-world flight test. (a) Video snapshot of the 10th second of the real-world experiments. (b) Video snapshot of the 30th second of the real-world experiments. (c) Video snapshot of the 50th second of the real-world experiments.

the flight phase. The attitude angle curve of the UAV during the flight test is shown in Fig. 17, where it can be seen that there were certain jumps and fluctuations, which were mainly caused by the wind disturbances in the outside flight environment. The vehicle was robust and adaptive enough to resist the disturbances and return to its desired movement state in the changing flight environment. Table IV summarizes the RMSE value of two controllers. From Table IV, it can be observed that compared with the CADRC controller, our proposed RL-based controller has higher control accuracy in the real-world scenario under wind environments. Through the above relevant flight experiments, the high precision and high effectiveness of tracking performance as well as strong robustness of the proposed control method were confirmed.

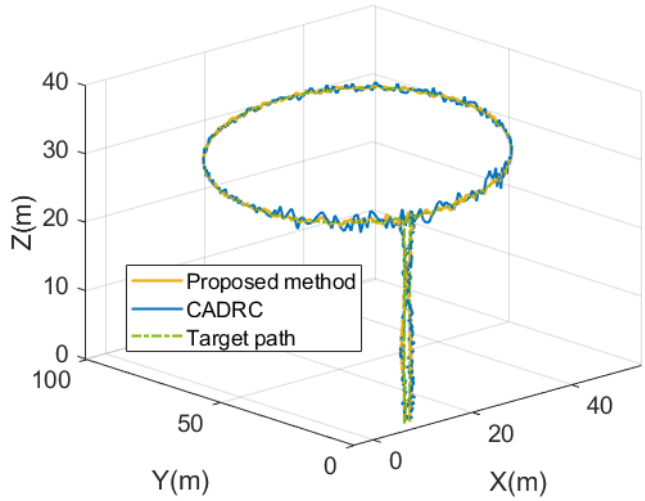

Fig. 15. 3-D trajectory of tracking control task in real-world flight test.

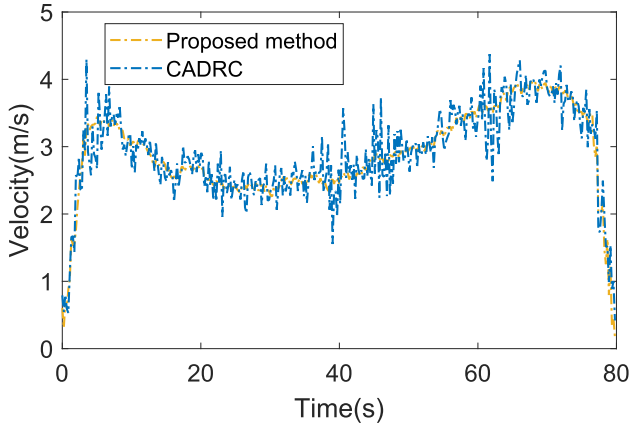

Fig. 16. Velocity control results in real-world flight test.

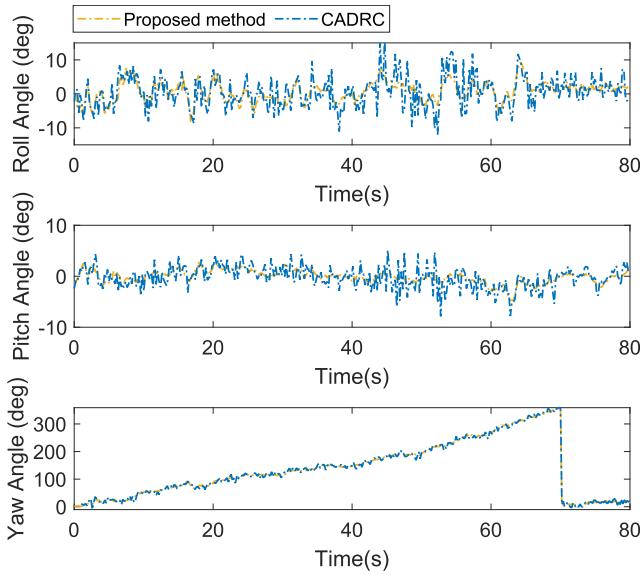

Fig. 17. Attitude control results in real-world flight test.

IV. Conclusion

This article proposes an innovative reinforcement-learning-based UAV tracking control algorithm in a dynamic environment, which can handle various types of wind disturbances

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during the tracking mission. The main goal of this study is to make the controller adjustable and robust when there are changes and disturbances in a flight environment. The PR mechanism is designed to encourage proper exploration in the learning phase, which reduces the imbalances between the new and existing knowledge. The SW method is used to enable an agent to adapt to various environments better. The random wind disturbance signal has been implemented on numerical dynamic environment in learning process for improving the robustness of the controller. Furthermore, a novel reward function is designed, which can increase the learning efficiency. Finally, a set of experiments, including the numerical simulation, HITL experiments, and the real-world flight test, are performed to verify the capacity and robustness of the proposed method under changing dynamic environments. The results demonstrate that the proposed RL-based control algorithm is effective and practical. In current research work, we focused on the control accuracy of the proposed method under wind disturbances and did not consider the collision avoidance function in multiobstacles' existing environments. In the future work, we will endeavor to investigate the collision avoidance problem of UAV in complex environments via external sensor system including Lidar and visual navigation module.

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